

MUHAMAD YOGA IBRAHIM

AI Engineer • Applied ML, LLM Integration & Data Systems

Bogor, Indonesia | +62 812-9235-8420 | yoga.ibh205@gmail.com

[linkedin.com/in/muhamadyogaibra](https://www.linkedin.com/in/muhamadyogaibra) | github.com/yogaibhh | drive.google.com/drive/folders/1ynH4sqZyUr4hk4cNcNFs-mrDQdosN2F

SUMMARY

AI/ML-focused engineer with hands-on experience training and deploying an edge-inference classification model (PyTorch, TensorFlow Lite), integrating multiple LLM providers (OpenAI, Groq, OpenRouter, Deepgram) into production tooling, and building agentic data workflows with the Model Context Protocol (MCP). Currently own full-stack scope (frontend, backend, database) in an AI Native engineering track: shipped a standardized full-stack boilerplate that replaced a fragmented, bug-prone internal dashboard system, and researching further applied-AI capabilities in multi-format data cleaning, document extraction, and geospatial code generation. Background in data engineering, including Python ETL pipelines, Elasticsearch, and PostgreSQL, provides a practical foundation for building and maintaining the data layer that AI systems depend on.

SKILLS

AI / ML: LLM API Integration (OpenAI, Groq, OpenRouter, Deepgram), Model Context Protocol (MCP), PyTorch, TensorFlow Lite, Edge AI Inference, Predictive Modeling, Agentic Workflows, AI Code Generation

Full-Stack & Data Engineering: Python (Pandas, NumPy), SQL, PostgreSQL, Elasticsearch, Lucene Queries, ETL Pipeline Design, Frontend/Backend Standardization

Systems & Frontend: JavaScript (React), Dart (Flutter), Electron

Geospatial: Google Earth Engine (GEE), Mapbox GL, ArcGIS

EXPERIENCE

AI Native Engineer PT Pixel Digital (PT Ebdesk Teknologi), Tangerang	Jun 2026 – Present
- Own full-stack scope (frontend, backend, database) building applied-AI “skills” that automate internal engineering workflows, moving beyond the prior data-scientist-only remit. - Migrating a fragmented internal BI-style dashboard system, originally built as multiple separate applications where updates frequently broke features across several apps at once, into a standardized full-stack boilerplate (consistent backend and frontend structure) built through AI-assisted code generation. Bug handling is now faster and more consistent across dashboards, removing a structural limitation of the legacy ERP setup. [needs: concrete before/after time figures, # of apps migrated] - Researching additional applied-AI capabilities: multi-format data cleaning (documents, spreadsheets, video, audio, satellite imagery), handwritten-document extraction from scanned PDFs, and Google Earth Engine code generation from its satellite catalog. [research stage, not yet in production]	
Data Scientist PT Pixel Digital (PT Ebdesk Teknologi), Tangerang	Oct 2025 – Jun 2026
- Engineered Python-based ETL pipelines to parse and normalize high-volume unstructured data feeding the National Stability Index and NPI predictive models, maintaining data integrity across the ingestion pipeline. - Built React-based geospatial monitoring dashboards backed by Elasticsearch/Lucene queries and Mapbox GL, rendering real-time spatial and network-flow (Sankey) data for security operations use cases. - Developed a fire-risk prediction model (Karhutla) using Google Earth Engine and NASA POWER meteorological data to support environmental risk monitoring. - Prototyped AI-agent-assisted dashboard and template generation for internal ERP tooling, an early version of the full-stack standardization work later formalized as a dedicated skill in the AI Native role above.	
Data Analyst PT Pixel Digital (PT Ebdesk Teknologi), Tangerang	May 2025 – Oct 2025
- Integrated multi-agent simulation outputs into PostgreSQL, connecting raw simulation logs to downstream visualization layers. - Built dashboards tracking trade patterns and geopolitical trends for stakeholder use.	
Data Analyst Anugerah Wisesa Selaras (weathermod.id), Bandung	Oct 2024 – Dec 2024
- Processed and analyzed 480+ hours of weather data in Python (Google Colab) and Excel, automating cleaning and visualization workflows and cutting reporting time by 40%. - Built automated data processing workflows to clean, transform, and analyze large meteorological datasets.	
Research Assistant PIAREA Environment & Technology, Bogor	Sep 2023 – Oct 2023

- Processed 175,000+ hourly weather data points (10 years, rainfall & temperature) in Excel and compiled the 2023 DKI Jakarta Air Quality Monitoring Map Album across 20+ monitoring locations.

PROJECTS

Cloud Seeding Hunter | *Internal Project (Confidential)*

- Trained and deployed a lightweight classification model (PyTorch, TensorFlow Lite) for offline, on-device edge inference, enabling field operators in remote areas to run predictions without connectivity.
- Built the end-to-end Flutter mobile app used to analyze meteorological data and support cloud-seeding operations, including an interactive geospatial tracking dashboard for the field engineering team.

AI Screen Reader | *Internal Productivity Tool*

- Built an Electron/React desktop app integrating multiple LLM providers (Deepgram, Groq, OpenRouter) to read and interpret on-screen visual context automatically.
- Managed API credential handling and environment configuration within a compiled, cross-platform desktop executable.

MCP AI Data Analyst | *Mar 2025 – Apr 2025*

- Configured Model Context Protocol (MCP) to connect a local PostgreSQL instance for real-time, natural-language querying of production tables.
- Scoped MCP to a strict read-only role, sandboxing all AI operations to prevent unintended writes or schema changes.

EDUCATION

B.S., Applied Meteorology, GPA 3.67 / 4.00 | IPB University, Bogor 2020 – 2024

- Relevant coursework: Data Processing (R, Excel), GIS & Remote Sensing (ArcGIS, GEE), Statistical Analysis & Visualization (Minitab, Excel).
- Presented “AOD and PM Relationship in Jakarta (2020–2021)” at the Indonesian Aerosol Association Conference, 2024.

CERTIFICATIONS

- Working with the OpenAI API, DataCamp, Oct 2025
- Associate Business Analyst in SQL, DataCamp, May 2025
- Associate Data Analyst in SQL, DataCamp, Apr 2025
- Data Analyst, Generasi Gigih 3.0, GoTo Impact Foundation, Jun 2023 – Nov 2023